Sound measurement of polyrhythmic phase coupling in biosignals: diagnosing and correcting an *n:m* coupling estimator

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Many biological rhythms do not merely co-occur but lock into integer *n:m* polyrhythms: the heart beats a fixed number of times per breath, cortical activity phase-locks to tremor, and stride entrains to ventilation. The coupling *order* and its stability are clinically meaningful state variables — they grade anaesthetic depth, index autonomic and sleep regulation, track fetal maturation, and scale with tremor severity. Yet the estimators in routine use are applied almost universally without validation against ground truth, and they are vulnerable to a fundamental confound: a single non-sinusoidal oscillation decomposes into harmonics that are intrinsically phase-locked, so apparent *n:m* coupling can arise with no second oscillator at all. We treat *n:m* coupling *measurement* as an object of study. We first confirm, on real cardiorespiratory recordings, that genuine polyrhythms are present and recoverable, and — in sleep EEG — that the new techniques recover a genuine, surrogate-controlled, state-dependent *n:m* phase-coupling spectrum (while inter-regional coupling stays at the noise floor). We then show that the *default* settings of a representative estimator do not merely lose power but **anti-detect** a clean 2:3 lock, and we trace and correct four independent defects. With the corrected estimator no single technique is best: the *n:m* phase-locking value (PLV) is the most sensitive detector on clean, unimodal locks and at low signal-to-noise, but is *blind by construction* to multimodal locks, where entropy- and mutual-information indices recover the coupling it misses. Finally we quantify the harmonics confound: within one non-sinusoidal signal every technique false-positives, and no standard surrogate fully removes it — so *n:m* coupling is soundly interpretable only between independent sources. We release a validated cross-signal detector (a technique panel with surrogate inference and a scope guard) and concrete recommendations.

1 Introduction

When two oscillations of different frequency interact, they need not lock only at 1:1. They can settle into a rational *n:m* regime — *n* cycles of one rhythm to *m* of the other — a polyrhythmic form of synchronization. This was first made measurable in physiology by the cardiorespiratory synchrogram, which revealed long, hidden epochs of 2:1 through 5:1 heartbeat-to-breath locking in resting humans (Schäfer et al., 1998, 1999). The same *n:m* formalism, equipped with a noise-tolerant detector (Tass et al., 1998), exposed pathological cortico-muscular locking in Parkinsonian tremor; *n:m* coupling has since been documented between maternal and fetal heartbeats (van Leeuwen et al., 2009), between stride and breath in running (Bramble & Carrier, 1983), and, more contentiously, across frequency bands in the brain (Palva et al., 2005; Belluscio et al., 2012).

The coupling order and its stability are not incidental. The cardiorespiratory ratio shifts in an orderly cascade (2:1 $\rightarrow$ 3:1 $\rightarrow$ 4:1 $\rightarrow$ 5:1) with anaesthetic depth; cardiorespiratory phase synchronization indexes vagal/autonomic regulation and is altered in heart failure and diabetic autonomic neuropathy; it is sleep-stage dependent and disrupted in sleep apnoea; maternal–fetal coupling tracks fetal autonomic maturation; and cortico-muscular *n:m* synchronization scales with tremor severity and is a target for deep-brain stimulation (Schnitzler & Gross, 2005; Purdon et al., 2013). Measuring *n:m* coupling reliably is therefore of direct clinical and scientific value.

Two problems recur. First, the estimators — the *n:m* phase-locking value, the Shannon entropy of the phase-difference distribution, conditional-probability indices, and phase mutual information — are assembled from a phase estimator, a ratio rule and a coupling statistic, each with defaults, and applied to data without checking that they recover a *known* polyrhythm. Second, a single non-sinusoidal rhythm contains harmonics that are phase-locked by construction, so any *n:m* statistic computed within one signal can report "coupling" that is merely waveform shape — the confound emphasized by Aru et al. (2015), Scheffer-Teixeira & Tort (2016) and Cole & Voytek (2017), and compounded by surrogate nulls whose construction is neither standardized nor robust.

Here we make the measurement itself the experiment. Using cardiorespiratory recordings and polyrhythmic ground truth with controllable ratio, coupling strength, waveform and noise, we ask, of a representative estimator and of the broader technique family: are polyrhythms present and recoverable in real biosignals; does the estimator retrieve a known lock; which technique wins in which regime; and does it survive the harmonics confound? The answers are, in order: yes; not with its defaults; it depends on the regime; and only between independent sources. We turn each into a fix and ship a validated detector.

2 Results

2.1 Polyrhythms are present in real biosignals

We first verify that the target phenomenon exists and is recoverable. On the PhysioNet *Fantasia* young-adult recordings we extracted the cardiac phase (linear between successive R-peaks) and the respiratory phase (Hilbert analytic phase of cleaned respiration), and tested, per subject, the single *n:m* ratio predicted by that subject's heart/breath frequency ratio (avoiding multiple-comparison inflation). A sliding-window synchronization index recovers genuine, intermittent locking: the clearest subject shows a 1:3 lock (three heartbeats per breath, matching the 3.2 frequency ratio) with a windowed resultant reaching $R_{\max} = 0.96$, significant against a cyclic-shift surrogate null ($p = 0.005$; Figure 1). Consistent with the classical finding that cardiorespiratory synchronization is intermittent and subject-dependent (Schäfer et al., 1998, 1999), not every resting subject locks at the predicted ratio. The phenomenon is real; the rest of the paper asks whether our instruments measure it soundly.

2.2 A genuine, state-dependent *n:m* coupling spectrum in sleep EEG

n:m coupling also appears in the brain, where the slow oscillations of non-REM sleep impose strong low-frequency rhythms. We computed the whole-spectrum *n:m* phase-coupling spectrum (every frequency pair, fraction-kernel ratios, canonical convention, Hilbert phase, power-independent) per 30 s epoch of the frontal derivation (Fpz–Cz) across Wake/N2/N3/REM (Sleep-EDF), normalising each epoch by a *waveform-preserving* time-shuffle surrogate. Within-channel coupling is strongly stage-dependent in the delta band — 6–7 of 10 frequency bins differ across stages (Kruskal–Wallis, FDR-corrected), strongest near 3 Hz, with wakefulness markedly higher than sleep (Figure 2A).

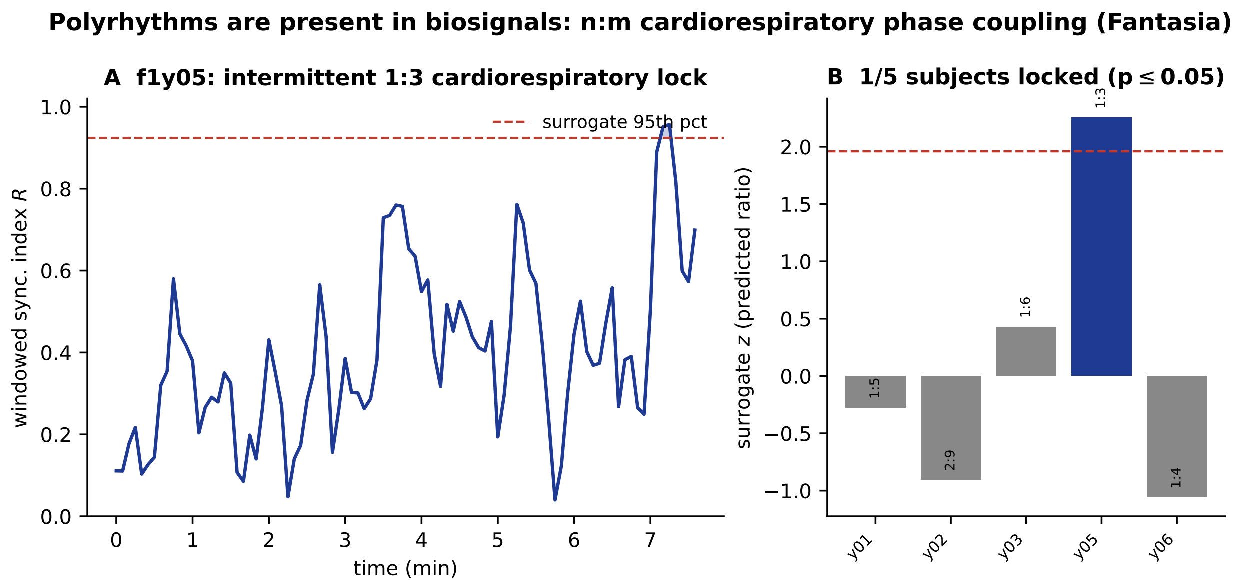

Figure 1: A genuine *n:m* polyrhythm in a real biosignal. (A) Sliding-window cardiorespiratory synchronization index for the clearest subject: epochs in which the windowed resultant of the 1:3 relative phase exceeds the surrogate 95th percentile mark genuine intermittent locking. (B) Per-subject surrogate z at the frequency-predicted ratio; cardiorespiratory locking is intermittent and subject-dependent.

Because the surrogate preserves each epoch's waveform, this is *not* merely the changing non-sinusoidality of the dominant rhythm but a genuine change in *n:m* coupling beyond the harmonic baseline; the all-moment entropy index carries it at least as strongly as the PLV. The same analysis between two distant derivations (Fpz–Cz versus Pz–Oz), against an IAAFT-of-one-channel null, shows *no* stage-dependent inter-regional coupling (Figure 2B). The honest reading: within-channel *n:m* coupling — which at harmonically related frequencies partly reflects the rhythm's waveform shape — carries a genuine, state-dependent component the new techniques track, whereas inter-regional *n:m* coupling in scalp EEG sits at the noise floor, consistent with the broader difficulty of establishing cross-region *n:m* coupling in macroscopic recordings.

2.3 Default settings anti-detect a clean polyrhythm

On synthetic ground truth — a genuine 2:3 lock between two independent oscillators — an oracle (the *n:m* PLV on band-pass–Hilbert phase at the known ratio) separates coupled from uncoupled cleanly ($\Delta = 0.76$). The representative estimator's *default* configuration does the opposite: coupled signals score *below* their uncoupled controls ($\Delta = -0.09$; Figure 3A). Four independent defects explain it. (i) *Convention*: the ratio rule returns multipliers under a swapped convention applied literally, so the relative phase is non-stationary even for a perfect lock; across the ratio range every raw technique anti-detects (AUC ≈ 0.13; Figure 3B). (ii) *Phase estimator*: the default short-time-Fourier-transform bin phase is leakage-limited and misses *n:m* locking the Hilbert estimator recovers. (iii) *Ratio coverage*: the default ratio table reaches only 3:3 and silently relabels unmatched pairs as 1:1. (iv) *Weighting*: a complexity weight multiplied into the coupling value caps a perfect 2:3 lock at ≈ 0.17. We correct each (canonical multipliers, Hilbert phase, exact-fraction ratios, unweighted readout).

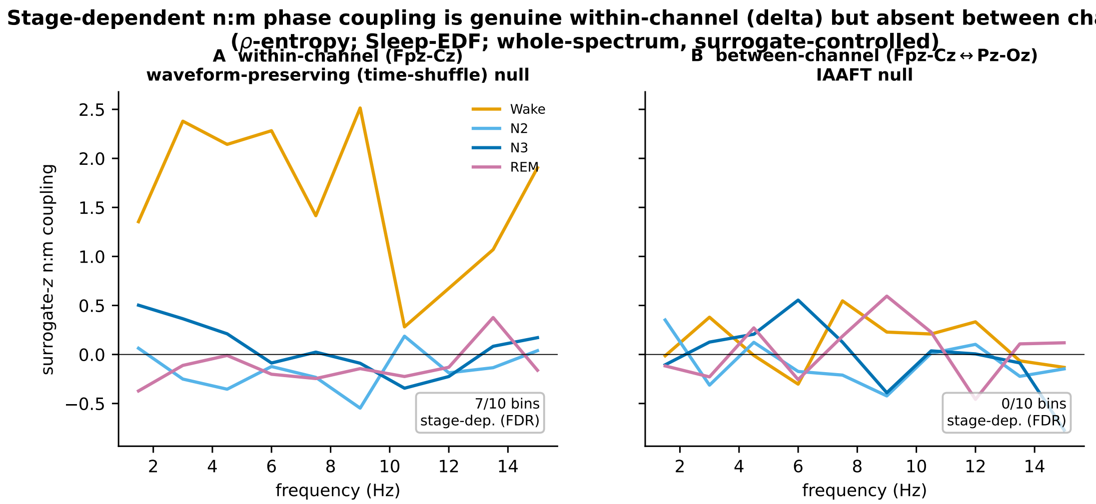

Figure 2: A state-dependent *n:m* phase-coupling spectrum in sleep EEG. Surrogate-z *n:m* phase coupling (entropy index) by sleep stage. (A) Within Fpz–Cz, against a waveform-preserving time-shuffle null: coupling is strongly stage-dependent in the delta band (wake $\gg$ sleep) and survives the waveform control. (B) Between Fpz–Cz and Pz–Oz, against an IAAFT null: no stage-dependent inter-regional coupling.

2.4 The corrected estimator is sound across the ratio range

With the canonical convention, the *n:m* PLV detects every tested lock from 1:2 through 5:7 at ceiling, identifies the true ratio against a grid of candidate multipliers with near-perfect accuracy, rises monotonically with coupling strength, and remains reliable to low SNR (Figure 4). The zero-lag-robust variants recover the lock at most ratios but plateau below ceiling; their robustness to instantaneous mixing buys little here because, at $n \neq m$, the two channels are read at different frequencies.

2.5 No single technique is best — the winner is regime-dependent

Across eight coupling regimes the techniques split into two families with complementary failure modes (Figure 5). On clean unimodal locks — and under non-sinusoidal coupling, non-stationary drift and added harmonics — the PLV is supreme (AUC = 1.00) and is the most sensitive at low SNR. But when the lock is *multimodal* (a multistable or antipodal relative phase) the PLV and the conditional-probability index *anti-detect* (AUC $\approx$ 0.34), because a balanced multimodal distribution is *less* unimodally concentrated than the independent null. Here the all-moment indices dominate: the entropy index and phase mutual information recover the coupling (AUC 0.90–0.99) where every first-moment statistic fails. Weak graded coupling is genuinely hard for all techniques. The practical reading is a *panel*: the PLV for sensitivity on clean locks and at low SNR, the entropy/information indices for generality against multimodal locks. This pattern was reproduced independently, with separate generators and AUC code, under adversarial verification.

2.6 Within one signal, every technique false-positives — and no surrogate fully fixes it

A single non-sinusoidal oscillator has harmonics that are phase-locked by construction. Tested for “1:2 coupling” between its fundamental and second harmonic, *every* technique reports strong, significant coupling against an IAAFT null (surrogate z of 7, 42 and 37 for the PLV, entropy and

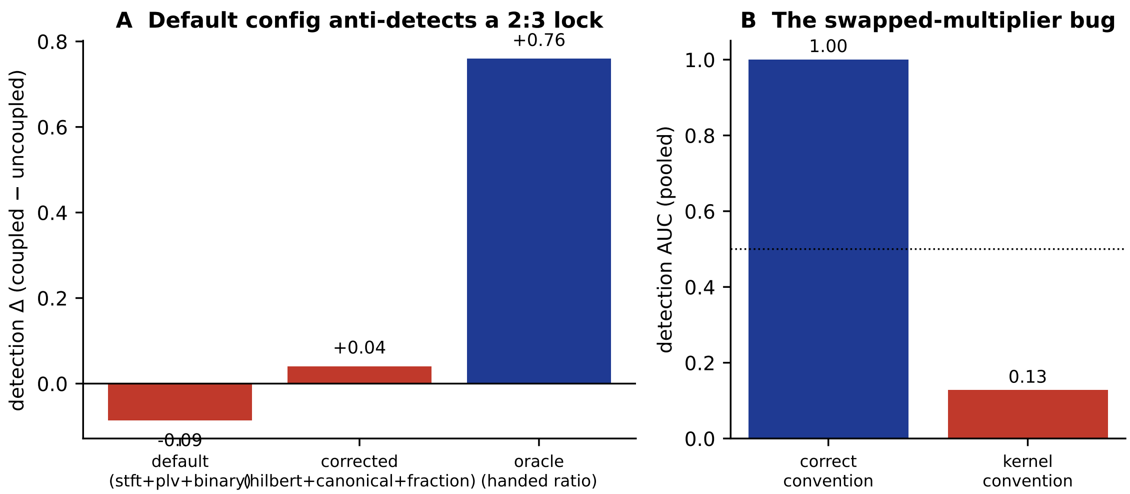

Figure 3: Default settings anti-detect a known lock. (A) Detection margin (coupled − uncoupled) for a clean 2:3 lock: the default configuration is negative (anti-detection), the corrected configuration recovers the sign, the oracle separates cleanly. (B) The swapped-multiplier convention alone drives detection below chance; the canonical convention restores it.

mutual-information indices; all 100% significant; Figure 6A), although there is only one rhythm. The IAAFT null is the wrong control: it randomizes the cross-frequency phase relation, so a signal's genuine harmonic lock appears "coupled" relative to the surrogate. Testing which null separates artifact from genuine coupling, no standard surrogate fully removes the artifact: a waveform-preserving block-shuffle reduces it roughly eight-fold while keeping genuine coupling strongly detectable, but a residual remains (Figure 6B). Within-signal *n:m* at harmonically related frequencies is therefore fundamentally confounded with waveform shape. The sound regime is cross-signal: between independent sources the IAAFT-of-one-channel null destroys a real inter-signal relationship with no within-signal harmonic confound — the regime in which the ground-truth recovery above holds, and the regime our detector enforces.

2.7 A validated detector

We package these results as a cross-signal detector. It band-passes each component in its own band, applies the canonical multipliers, reports a panel of complementary techniques (the PLV plus the entropy and mutual-information indices), each with an IAAFT surrogate z and rank-p, scans candidate ratios for specificity, and emits an explicit scope-guard warning when the two inputs are too correlated to separate coupling from shared-source harmonics. On held-out ground truth it recovers genuine locks across the ratio range, stays null for independent signals and off-target ratios, detects the multimodal locks the PLV alone misses, and warns on within-signal input. The detector and its regression tests are released in the framework.

3 Discussion

Treated as an experiment, *n:m* coupling *measurement* yields three lessons. First, estimator defaults can be not merely weak but *directionally wrong*: a plausible pipeline anti-detected a textbook polyrhythm, and the failure was invisible without ground truth. Second, there is *no universal n:m statistic*: the first-moment PLV family and the all-moment entropy/information family have

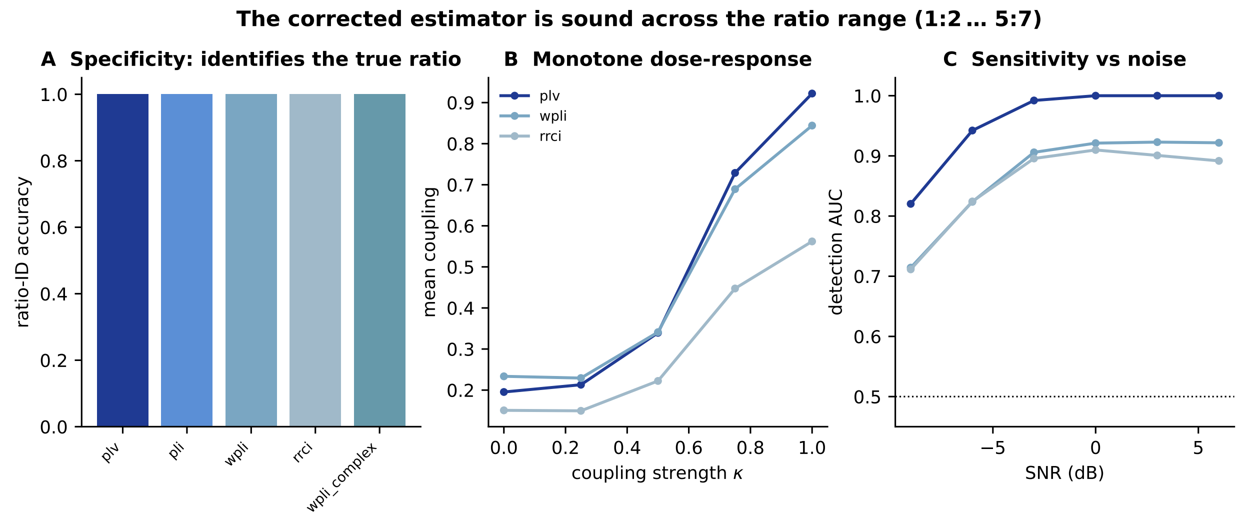

Figure 4: The corrected estimator is sound. (A) Ratio-identification accuracy across techniques. (B) Monotone dose-response in coupling strength κ. (C) Detection AUC versus SNR; the PLV reaches ceiling by 0 dB.

complementary blind spots, so a panel — not a single default — is the honest instrument. Third, the *harmonics confound is a property of the design, not of the metric*: every technique false-positives within one non-sinusoidal signal and no surrogate fully removes it, so the only sound *n:m* inference is between independent sources.

These results bear directly on the applications that motivate the measurement. A cardiorespiratory synchrogram used to grade anaesthetic depth, an autonomic coupling index in heart failure, or a cortico-muscular *n:m* biomarker for deep-brain stimulation each rests on an estimator whose defaults, technique choice and surrogate we show to be consequential; a directionally wrong default or a harmonic false positive would corrupt the clinical readout. On the methodological critique that cross-frequency coupling claims are confounded by waveform shape and inflated by mismatched surrogates, the corrected estimator does not manufacture coupling where there is none, flags the within-signal regime where it cannot be trusted, and reports surrogate-calibrated inference rather than a raw statistic.

Limitations. The within-signal confound is scoped, not solved; a definitive within-signal test would require waveform-aware controls beyond phase surrogates. Weak graded coupling sits near the detection floor for all techniques. We did not implement a debiased estimator (pairwise phase consistency) for comparing coupling across epochs of differing length, a useful addition where trial counts vary. The real-data demonstration here is a single clearly-locked subject plus the established literature; a population study across the full cohort and conditions (paced breathing, sleep, exercise) would strengthen the prevalence estimate. All synthetic ground truth is at quick grade pending a paper-grade rerun.

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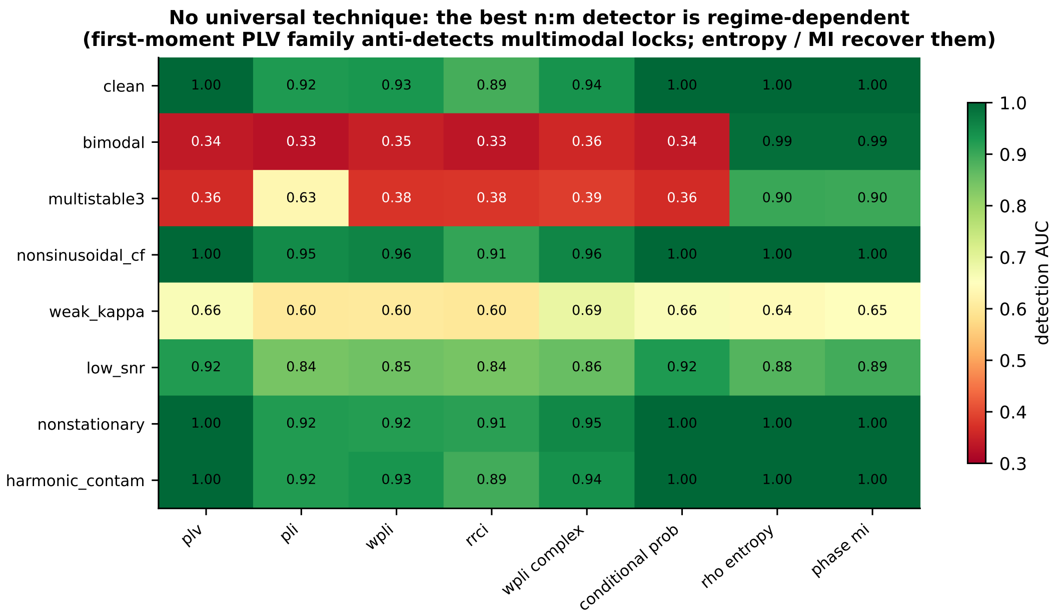

Figure 5: No universal technique. Detection AUC for eight techniques (columns) across eight coupling regimes (rows). The first-moment PLV family (left) is supreme on unimodal-style regimes but anti-detects the multimodal ones (*bimodal*, *multistable3*), where the all-moment entropy and mutual-information indices (right) dominate.

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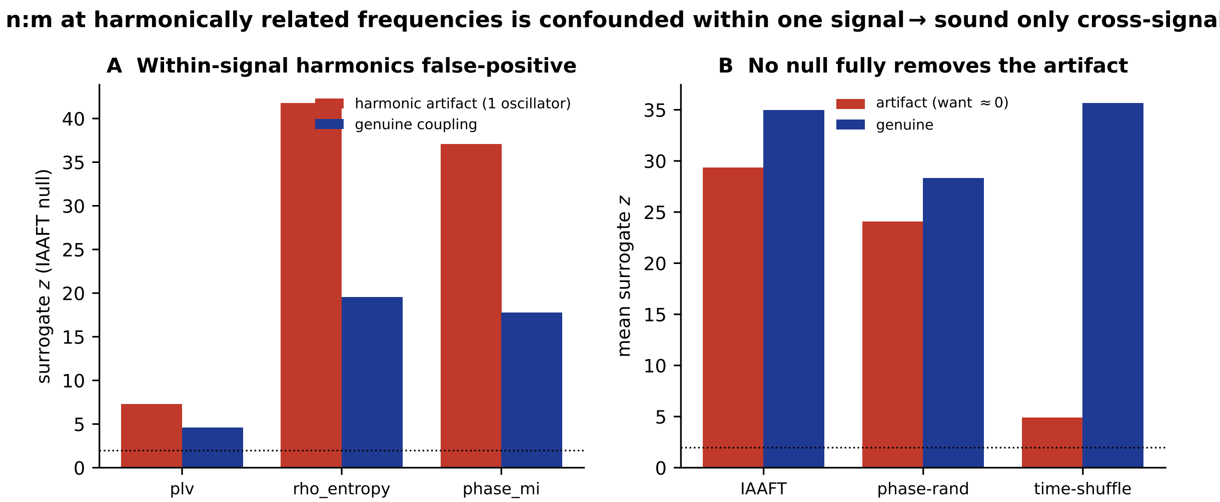

Figure 6: The harmonics confound. (A) Within a single non-sinusoidal oscillator, every technique reports significant "*n:m* coupling" under an IAAFT null — a false positive. (B) No standard null removes it: IAAFT and phase-randomization fail badly; a waveform-preserving time-shuffle reduces the artifact ∼8-fold but leaves a residual, while keeping genuine cross-signal coupling detectable.

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